

Лабораторная работа №6

Имитационное моделирование

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Раздел 1

Информация

- Ибрахим Мохсейн Алькамаль
- Российский университет дружбы народов

Раздел 2

Цель работы

- Изучить реализацию эпидемиологической модели SIR с использованием сетей Петри.
- Выполнить детерминированное и стохастическое моделирование.
- Исследовать влияние коэффициента заражения β .
- Сравнить полученные динамики модели.

Раздел 3

Выполнение лабораторной работы

Подготовка рабочего проекта

- Создан и активирован проект lab06/project.
- Использован скрипт setup_project.jl.
- Сформировано проектное окружение Julia.
- Установлены необходимые зависимости.

```
alkanal@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06$
julia setup_project.jl
Resolving package versions...
Project No packages added to or removed from `~/julia/environments/v1.12/Project.toml`
Manifest No packages added to or removed from `~/julia/environments/v1.12/Manifest.toml`
Activating new project at `~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project`
Resolving package versions...
Updating `~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/Project.toml`
[534d3b9d] + DrWatson v2.18.1
Updating `~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/Manifest.toml`
[9b6fb165] + ChunkCodeCore v1.8.2
[4c8bbe4] + ChunkCodeLibZLib v1.1.8
[55437552] + ChunkCodeLibZstd v1.8.8
[634d3b9d] + DrWatson v2.18.1
[5789e2e8] + FileIO v1.28.8
[876d061b] + HashArrayMappedTries v0.2.8
[833835bb] + JLD2 v0.6.6
[692b3bcd] + JLLWrappers v1.8.0
[1914dd2f] + MacroTools v0.5.16
[bac558e1] + OrderedCollections v2.0.1
[aea7be01] + PrecompileTools v1.3.4
[21216c8a] + Preferences v1.5.2
[ae028912] + Requires v1.3.1
[76586755] + ScopedValues v1.6.2
[6c6a2e73] + Scratch v1.3.8
[3a884ed8] + UnPack v1.8.2
[3161d3a3] + Zstd_jll v1.5.7+1
[8d4e04c3] + Arpack_jll v1.1.2
```

Рисунок 1: Создание рабочего проекта и установка зависимостей Julia

Проверка проектного окружения

- Проект запущен командой `julia --project=..`
- Состав окружения проверен с помощью `Pkg.status()`.
- Проверено наличие необходимых пакетов.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=..
Documentation: https://docs.julialang.org
Type "?" for help, "}" for Pkg help.
Version 1.12.1 (2025-10-17)
Fedora 44 build

julia> using Pkg

julia> Pkg.status()
Status ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/Project.toml
[46ada45e] Agents v7.0.3
[4f99eebe] AlgebraicPetri v0.8.11
[6e4b80f9] BenchmarkTools v1.8.0
[a134a8b2] BlackBoxOptim v0.8.12
[336ed88f] CSV v0.10.17
[X 134e6e36] Catlab v0.14.17
[a03c6ff0] DataFrames v1.8.2
[9e46a032] DifferentialEquations v8.1.1
[31c24e18] Distributions v0.25.131
[634d3b8d] DrWatson v2.19.1
[78731f75] IJulia v1.34.4
[033835bb] JLD2 v0.6.6
[98b081ad] Literate v2.21.0
[1dea7af3] OrdinaryDiffEq v7.8.1
[91a5bcdd] Plots v1.41.7
[d7167be5] Quarto v1.0.8
[2913bdd2] StatsBase v0.34.13
[f3b207a7] StatsPlots v0.15.8
[ada2cc78] Dates v1.11.8
Info Packages marked with ~ and X have new versions available. Those with ~ may be upgradable, but those with X are restricted by compatibility constraints from upgrading. To
use `status --outdated`

julia>
```

Рисунок 2: Проверка установленных пакетов проектного окружения Julia

Базовый прогон модели SIR

- Созданы модуль `SIRPetri.jl` и скрипт `sirpetri_run.jl`.
- Выполнен базовый прогон модели SIR.
- Результаты сохранены в `sir_det.csv` и `sir_stoch.csv`.
- Созданы графики `sir_det_dynamics.png` и `sir_stoch_dynamics.png`.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano src/SIRPetri.jl
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_run.jl
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_run.jl
Could not create decoration from factory! Running with no decorations.
Базовый прогон завершён. Результаты в data/ и plots/
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls plots
sir_det_dynamics.png  sir_stoch_dynamics.png
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls data
exp_pro  exp_raw  sims  sir_det.csv  sir_stoch.csv
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 3: Выполнение базового прогона модели SIR и сохранение результатов

Преобразование базового эксперимента в литературный стиль

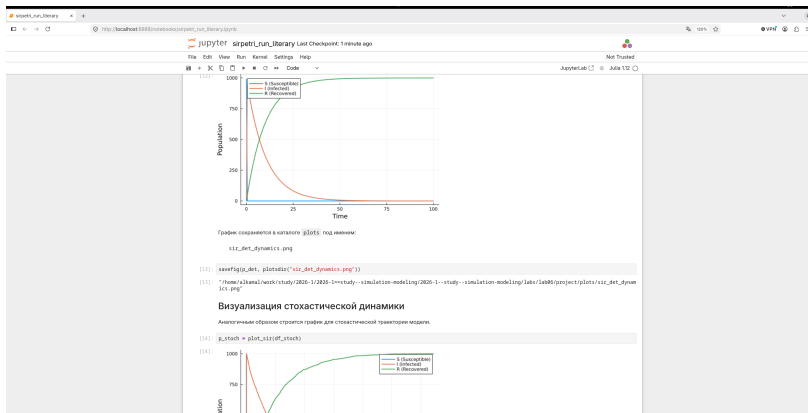
- Подготовлен файл `sirpetri_run_literary.jl`.
- Выполнен скрипт `tangle.jl`.
- Сформированы Julia-скрипт, документ Quarto и Jupyter Notebook.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ cp labs/lab01/project/scripts/tangle.jl labs/lab06/project/scripts/
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ cd labs/lab06/project/scripts/
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts$ ls
intro.jl sirpetri_run.jl sirpetri_run_literary.jl tangle.jl
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab06/project/scripts$ cd ..
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/tangle.jl scripts/sirpetri_run_
literary.jl
Генерация из: scripts/sirpetri_run_literary.jl
[ Info: generating plain script file from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run_literary.
jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run_literary/sirpetri_run_lit
erary.jl"
✓ Чистый скрипт: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run_literary/sirpetri_run_
literary.jl
[ Info: generating markdown page from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_run_literary/sirpetri_run_li
terary.qmd"
✓ Quarto: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_run_literary/sirpetri_run_litera
ry.qmd
[ Info: generating notebook from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_run_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_run_literary/sirpetri_run_li
terary.ipynb"
✓ Notebook: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_run_literary/sirpetri_run_li
terary.ipynb
Готово! Все файлы созданы.
```

Рисунок 4: Генерация Julia-скрипта, документа Quarto и Jupyter Notebook для модели SIR

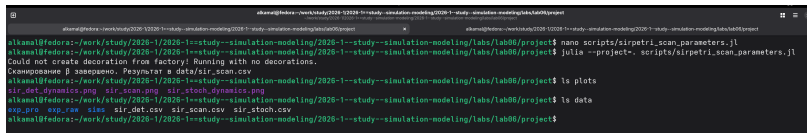
Выполнение базового эксперимента в Jupyter Notebook

- Выполнен `sirpetri_run_literary.ipynb`.
- Получена детерминированная динамика состояний S, I и R.
- Получена стохастическая динамика состояний S, I и R.
- Отображено изменение численности групп во времени.



Сканирование параметра заражения

- Выполнен скрипт `sirpetri_scan_parameters.jl`.
- Исследовано влияние коэффициента заражения β .
- Результаты сохранены в `data/sir_scan.csv`.
- График сохранён в `plots/sir_scan.png`.



```
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_scan_parameters.jl
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_scan_parameters.jl
Could not create decoration from factory! Running with no decorations.
Сканирование  $\beta$  завершено. Результат в data/sir_scan.csv
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls plots
sir_det_dynamics.png  sir_scan.png  sir_stoch_dynamics.png
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls data
exp_pro  exp_raw  sims  sir_det.csv  sir_scan.csv  sir_stoch.csv
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 6: Выполнение сканирования коэффициента заражения модели SIR

Преобразование сканирования параметров в литературный стиль

- Подготовлен файл `sirpetri_scan_parameters_literary.jl`.
- Выполнено преобразование с помощью `tangle.jl`.
- Сформированы Julia-скрипт, документ Quarto и Jupyter Notebook.

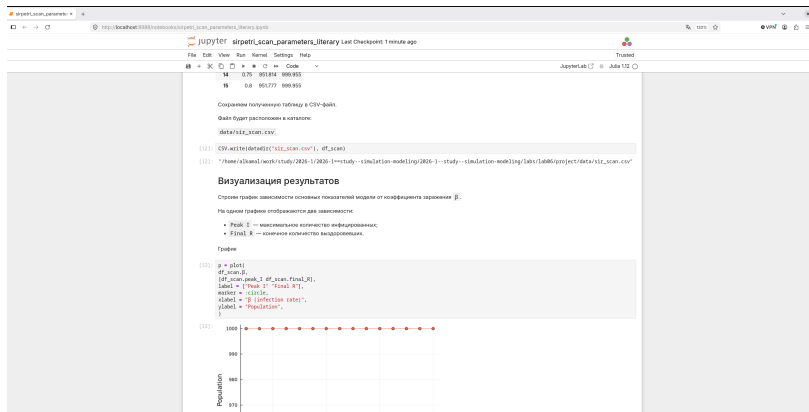
```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_scan_parameters_literary.jl
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/tangle.jl scripts/sirpetri_scan_parameters_literary.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_scan_parameters_literary.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_scan_parameters_literary/sirpetri_scan_parameters_literary.jl`
✓ Чистый скрипт: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_scan_parameters_literary/sirpetri_scan_parameters_literary.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_scan_parameters_literary.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_scan_parameters_literary/sirpetri_scan_parameters_literary.qmd`
✓ Quarto: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_scan_parameters_literary/sirpetri_scan_parameters_literary.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_scan_parameters_literary.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_scan_parameters_literary/sirpetri_scan_parameters_literary.ipynb`
✓ Notebook: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_scan_parameters_literary/sirpetri_scan_parameters_literary.ipynb

Готово! Все файлы созданы.
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 7: Генерация файлов литературной версии сканирования параметров модели SIR

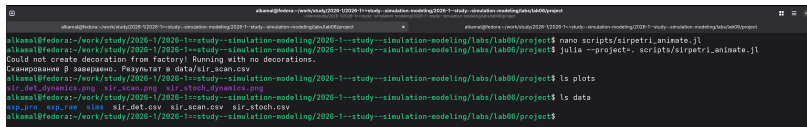
Анализ результатов сканирования в Jupyter Notebook

- Выполнен `sirpetri_scan_parameters_literary.ipynb`.
- Исследована зависимость `Peak I` от коэффициента β .
- Исследована зависимость `Final R` от коэффициента β .
- Результаты сохранены в `data/sir_scan.csv`.



Создание анимации модели SIR

- Подготовлен и выполнен скрипт `sirpetri_animate.jl`.
- Проверены результаты моделирования.
- Проверено содержимое каталогов `data` и `plots`.



```
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_animate.jl
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_animate.jl
Could not create decoration from factory! Running with no decorations.
Сканирование завершено. Результат в data/sir_scan.csv
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls plots
sir_det_dynamics.png  sir_scan.png  sir_stoch_dynamics.png
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls data
exp_pro  exp_raw  sim  sir_det.csv  sir_scan.csv  sir_stoch.csv
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 9: Выполнение скрипта визуализации и проверка результатов модели SIR

Преобразование скрипта визуализации в литературный стиль

- Подготовлен файл `sirpetri_animate_literary.jl`.
- Выполнено преобразование с помощью `tangle.jl`.
- Сформированы Julia-скрипт, документ Quarto и Jupyter Notebook.

```
alkamal@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_animate_literary.jl
alkamal@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project$ julia --project=. scripts/tangle.jl scripts/sirpetri_animate_literary.jl
Генерация из: scripts/sirpetri_animate_literary.jl
[ Info: generating plain script file from "~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/scripts/sirpetri_animate_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/scripts/sirpetri_animate_literary/sirpetri_animate_literary.jl"
✓ Чистый скрипт: /home/alkamal/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/scripts/sirpetri_animate_literary/sirpetri_animate_literary.jl
[ Info: generating markdown page from "~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/scripts/sirpetri_animate_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/markdown/sirpetri_animate_literary/sirpetri_animate_literary.qmd"
✓ Quarto: /home/alkamal/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/markdown/sirpetri_animate_literary/sirpetri_animate_literary.qmd
[ Info: generating notebook from "~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/scripts/sirpetri_animate_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/notebooks/sirpetri_animate_literary/sirpetri_animate_literary.ipynb"
✓ Notebook: /home/alkamal/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project/notebooks/sirpetri_animate_literary/sirpetri_animate_literary.ipynb

Готово! Все файлы созданы.
alkamal@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab06/project$
```

Рисунок 10: Генерация файлов литературной версии визуализации модели SIR

Выполнение литературной версии в Jupyter Notebook

- Выполнен `sirpetri_animate_literary.ipynb`.
- Получены результаты для β от 0.1 до 0.8.
- Рассчитаны показатели `peak_I` и `final_R`.
- Результаты сохранены в `data/sir_scan.csv`.
- Данные подготовлены для построения графика.

The screenshot shows a Jupyter Notebook window titled 'sirpetri_animate_literary'. The browser address bar shows 'http://localhost:8888/notebooks/sirpetri_animate_literary.ipynb'. The notebook interface includes a menu bar (File, Edit, View, Run, Kernel, Settings, Help) and a toolbar. The main area displays the following content:

```
[11]: df_scan = DataFrame(results)
```

Below this, a table view of the DataFrame is shown:

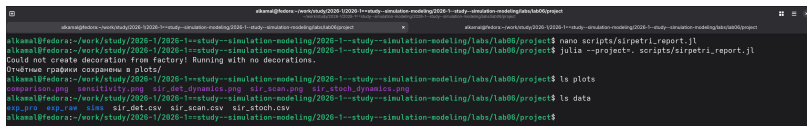
Row	β	peak_I	final_R
1	0.1	955.626	999.994
2	0.15	954.937	999.994
3	0.2	953.424	999.994
4	0.25	952.984	999.995
5	0.3	952.691	999.995
6	0.35	952.482	999.995
7	0.4	952.326	999.995
8	0.45	952.204	999.995
9	0.5	952.106	999.995
10	0.55	952.026	999.995
11	0.6	951.96	999.995
12	0.65	951.904	999.995
13	0.7	951.856	999.995
14	0.75	951.814	999.995
15	0.8	951.777	999.995

Below the table, there is a section titled 'Сохранение результатов' (Saving results) with the text: 'Полученная таблица сохраняется в CSV-файл: `data/sir_scan.csv`'. It also mentions that the `datafile` function from the `DrWatson` package automatically formats the path to the `data` directory of the project.

```
[12]: CSV.write(datafile("sir_scan.csv"), df_scan)
```

Формирование итогового отчёта

- Подготовлен и выполнен скрипт `sirpetri_report.jl`.
- Создан график `comparison.png`.
- Создан график `sensitivity.png`.
- Проверены результаты в каталогах `plots` и `data`.



```
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_report.jl
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_report.jl
Could not create decoration from factory! Running with no decorations.
Отчётные графики сохранены в plots/
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls plots
comparison.png  sensitivity.png  sir_det_dynamics.png  sir_scan.png  sir_stoch_dynamics.png
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ ls data
exp_pro  exp_raw  sias  sir_det.csv  sir_scan.csv  sir_stoch.csv
alkamal@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 12: Формирование отчётных графиков и проверка результатов моделирования SIR

Преобразование итогового отчёта в литературный стиль

- Подготовлен файл `sirpetri_report_literary.jl`.
- Выполнено преобразование с помощью `tangle.jl`.
- Сформированы Julia-скрипт, документ Quarto и Jupyter Notebook.

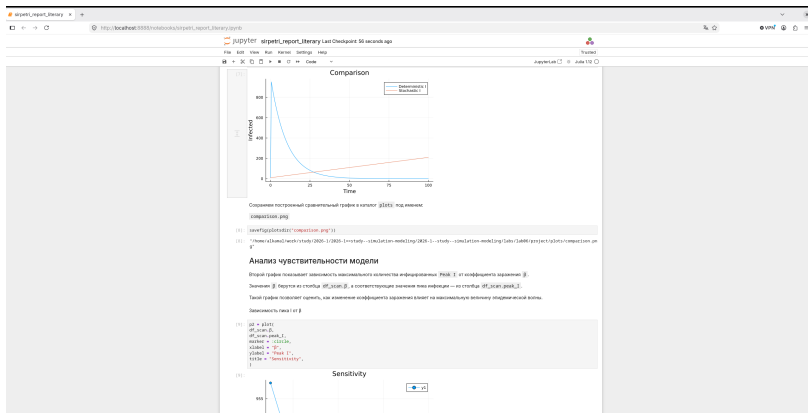
```
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ nano scripts/sirpetri_report_literary.jl
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/tangle.jl scripts/sirpetri_report_literary.jl
Генерация из: scripts/sirpetri_report_literary.jl
[ Info: generating plain script file from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_report_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_report_literary/sirpetri_report_literary.jl"
✓ Чистый скрипт: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_report_literary/sirpetri_report_literary.jl
[ Info: generating markdown page from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_report_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_report_literary/sirpetri_report_literary.qmd"
✓ Quarto: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/sirpetri_report_literary/sirpetri_report_literary.qmd
[ Info: generating notebook from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/sirpetri_report_literary.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_report_literary/sirpetri_report_literary.ipynb"
✓ Notebook: /home/alkamal/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/sirpetri_report_literary/sirpetri_report_literary.ipynb

Готово! Все файлы созданы.
alkamal@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$
```

Рисунок 13: Генерация файлов литературной версии итогового отчёта модели SIR

Анализ результатов в Jupyter Notebook

- Выполнен `sirpetri_report_literary.ipynb`.
- Сравнена детерминированная и стохастическая динамика инфицированных.
- Построена зависимость $Peak\ I$ от коэффициента заражения β .
- Проведён анализ чувствительности модели.



Раздел 4

Выводы

- Реализована модель SIR на основе сетей Петри.
- Выполнено детерминированное и стохастическое моделирование.
- Исследовано влияние коэффициента заражения β .
- Выполнено сравнение детерминированных и стохастических результатов.
- Проведён анализ чувствительности модели.
- Результаты сохранены в CSV-файлах и графиках.
- Литературные версии представлены в форматах Quarto и Jupyter Notebook.